LIGHTNING TALK #2

Credit Card Default Prediction

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Problem Statement

Can we predict **whether a customer will default** on their next credit card payment, using repayment history, credit limit, bill amounts, previous payments, and demographic information?

SUCCESS METRIC

ROC-AUC ≥ 0.75 and Recall ≥ 0.60 on the default class — missing an actual defaulter costs more than a false alarm.

Project Goals

BUSINESS GOALS

- Flag high-risk customers before they default
- Identify which behaviors and demographics drive risk
- Support credit-limit and collections decisions with data

MACHINE LEARNING GOALS

- Build a binary classification model
- Compare Logistic Regression, Random Forest, XGBoost
- Evaluate with Precision, Recall, F1, ROC-AUC, and Accuracy
- Tune each model with GridSearchCV

About the Data

30,000	**25**	**77.9%**	**22.1%**
customer records	columns — demographics, repayment history, bills, payments	no default	defaulted next month

Target: default.payment.next.month (1 = default, 0 = no default)

Moderate class imbalance — informs the choice of evaluation metrics and modeling strategy (class weights, stratified split).

Findings from Cleaning

MISSING VALUES

None found across all 25 columns.

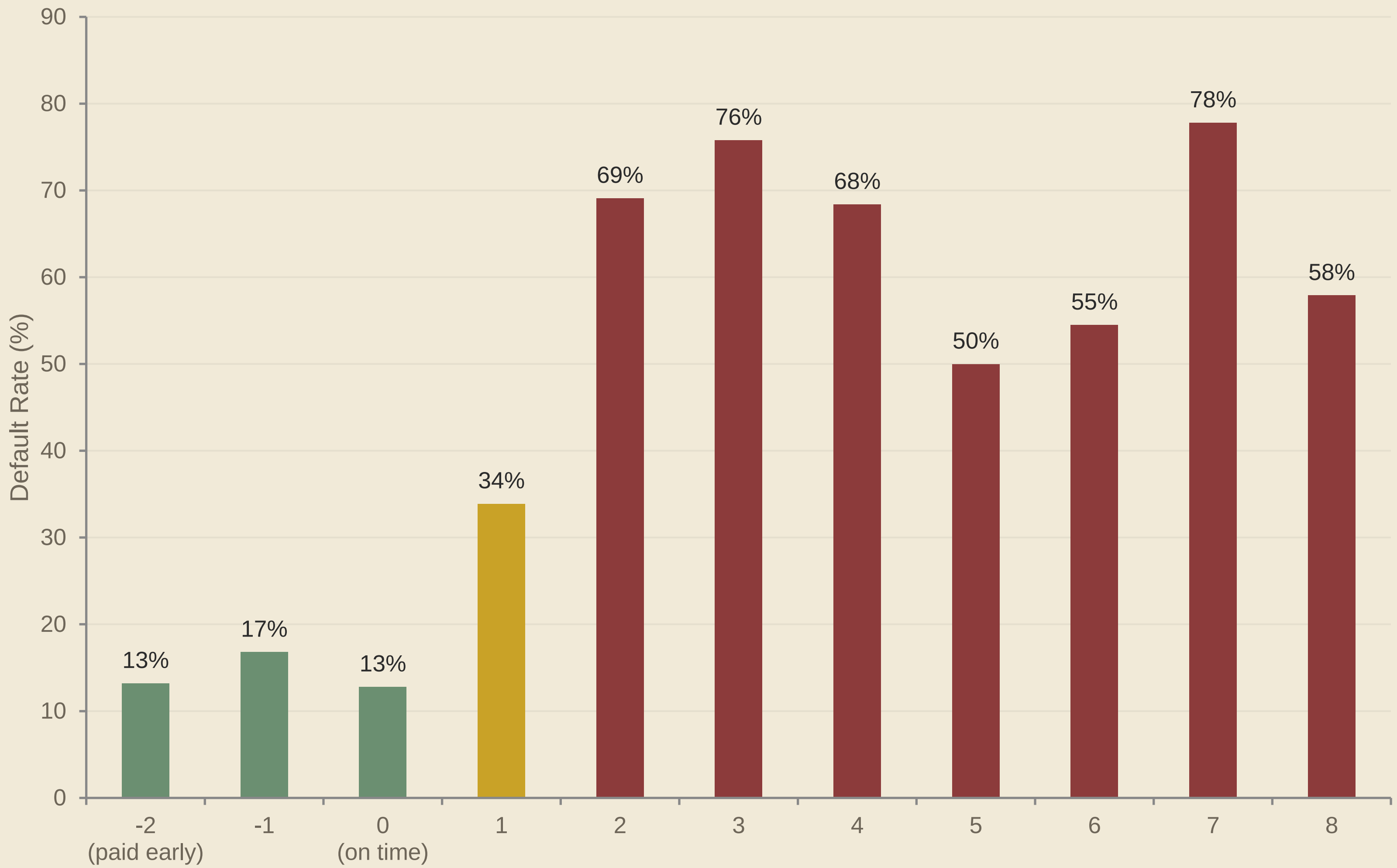
DATA QUALITY FIXES

EDUCATION and MARRIAGE contained undocumented / zero codes — recoded into an “other” category rather than dropped.

OUTLIERS — RETAINED

LIMIT_BAL, bill and payment amounts show high-value outliers. These are reasonable high-spending customers, not errors kept, may be transformed later for scale-sensitive models.

Strongest Predictor: Repayment Status



INSIGHT

Default rate stays low (13–17%) while payments are current, then climbs sharply once a customer is late:

1 month late → 34%

3+ months late → 76%+

PAY_0 (most recent repayment status) is the single strongest signal in the dataset.

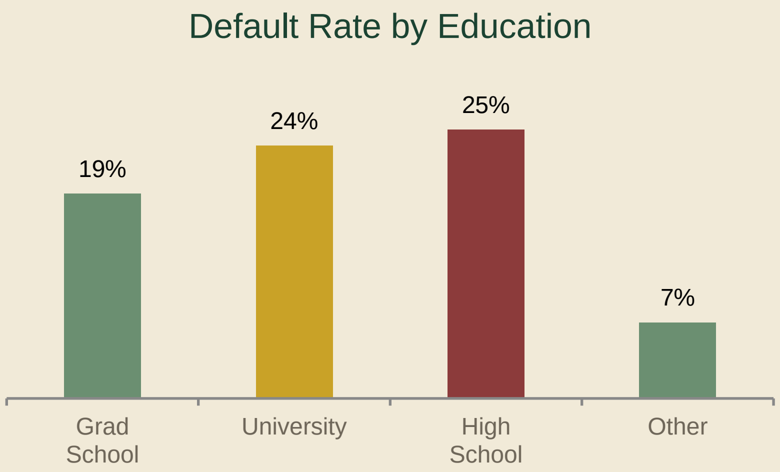
Credit Limit & Demographics

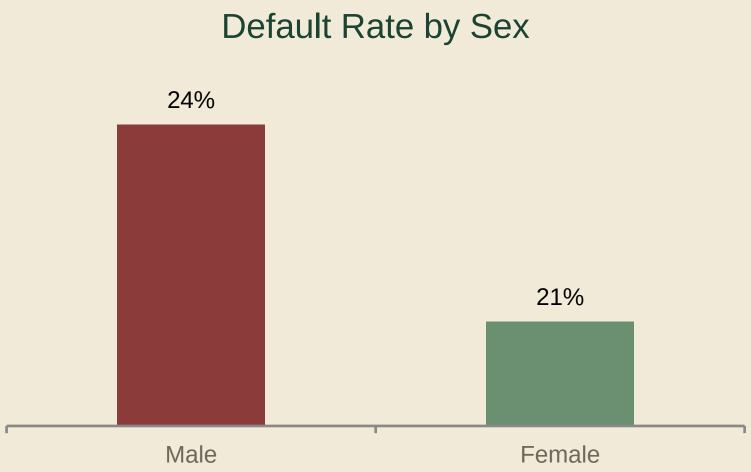
$178K

avg. credit limit
no default

$130K

avg. credit limit
defaulted





INSIGHT

Defaulters carry noticeably lower credit limits on average. Education shows a real gradient (19% → 25%) and sex a modest gap (3.4 pts), while marital status shows little difference (21–24% across groups) — demographics matter, but far less than repayment behavior.

Statistical Significance (Chi-Square)

Rank	Feature	Chi^2	p-value
1	**PAY_0**	5,366	< 0.001
2	**PAY_2**	3,474	< 0.001
3	**PAY_3**	2,622	< 0.001
4	**PAY_4**	2,341	< 0.001
5	**PAY_5**	2,198	< 0.001
6	**PAY_6**	1,887	< 0.001
7	**EDUCATION**	160	< 0.001
8	**SEX**	48	< 0.001
9	**MARRIAGE**	28	< 0.001

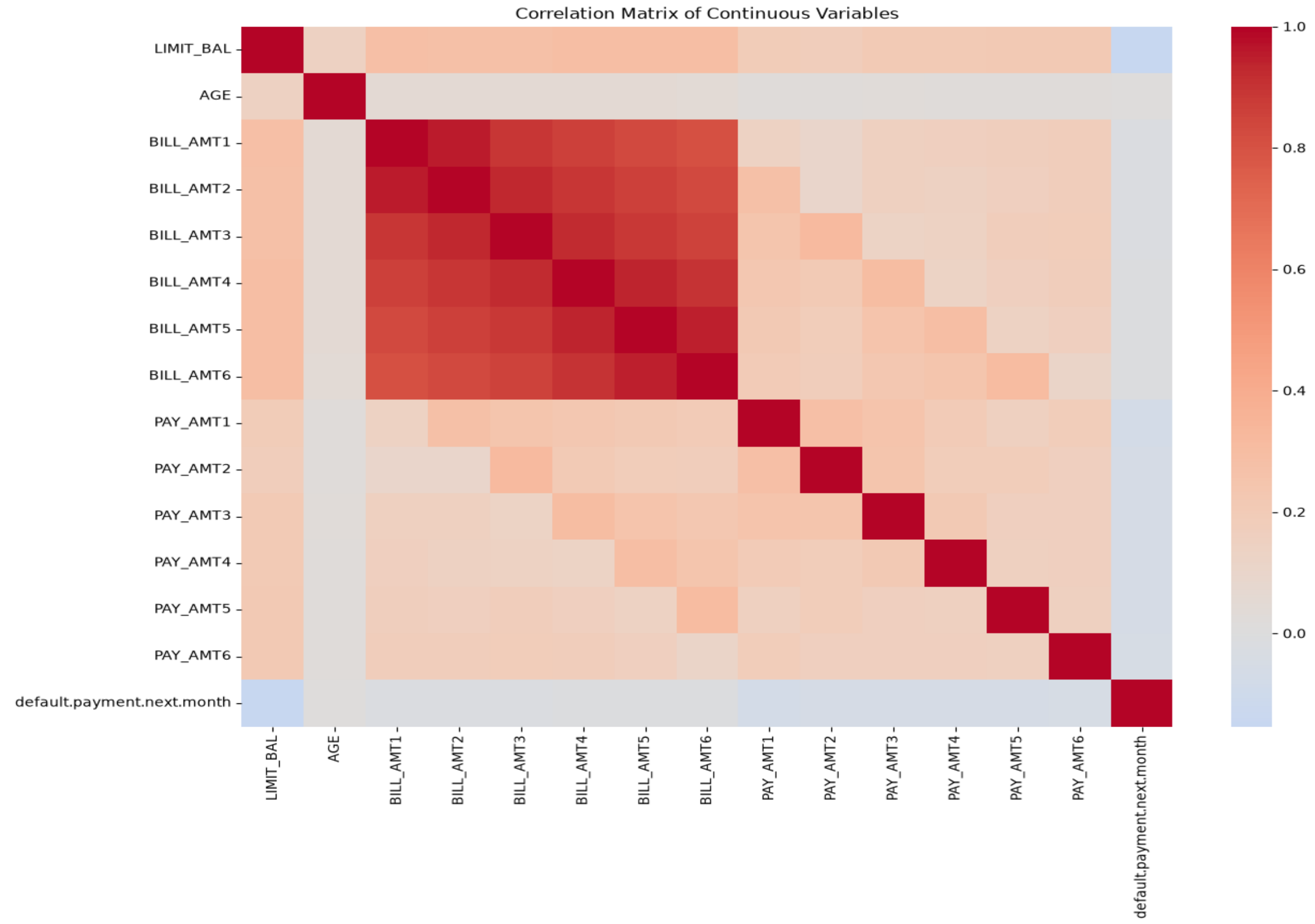
READING THE TABLE

All 9 categorical features are statistically significant ($p < 0.001$) — but significance isn't the same as strength.

Repayment status (PAY_x)
dwarfs demographics — Chi^2 in the thousands vs. tens-to-hundreds.

This ranking directly informs feature priority in modeling.

Correlation Among Continuous Features



INSIGHT

BILL_AMT1–6 are highly correlated with each other (> 0.9) — a multicollinearity risk for Logistic Regression.

Planned fix: consolidate into engineered features (e.g. average bill amount) to reduce redundancy before modeling.

LIMIT_BAL and AGE correlate only weakly with the target directly — their value comes through interaction with repayment behavior.

Planning

PREPROCESSING

- One-hot encode SEX / EDUCATION / MARRIAGE
- Keep PAY_x as ordinal features
- Standardize continuous features (Logistic Regression)
- Consider transforming highly skewed bill / payment amounts

FEATURE ENGINEERING

- Delayed-payment count
- Maximum repayment delay
- Average bill / payment amount
- Payment-to-bill & utilization ratios

MODEL APPROACH

- Logistic Regression (baseline)
- Random Forest
- XGBoost
- GridSearchCV tuning — train set only

Candidate Models

Model	Strength	Limitation
Logistic Regression	Simple, fast, interpretable — establishes a baseline	Assumes linear relationships; may underfit complex patterns
Random Forest	Captures nonlinear interactions; robust to outliers	Less interpretable; heavier compute than a single tree
XGBoost	Powerful boosting model that captures complex nonlinear patterns and often performs well on tabular data	More computationally intensive and requires careful hyperparameter tuning

Selection will weigh cross-validated Precision, Recall, F1, and ROC-AUC — with particular attention to Recall and F1 for the default class — before a final evaluation on the untouched test set.

Progress Report

CHALLENGES

- Multicollinearity among the six bill-amount columns
- Moderate class imbalance (22% default)
- Choosing which engineered features add signal vs. redundancy

NEXT STEPS

- Execute the preprocessing pipeline
- Engineer and validate new features
- Train and tune baseline models
- Evaluate on a held-out test set